Department: Software engineering

Course: Requirements Engineering

Title: Group Practical Assignment: SRS Documentation

Date: 13/10/2025

Group 1

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Software Requirements Specification (SRS)

Project: Takura Bid Ride-Hailing Platform

Document Information	Details
Title	Software Requirements Specification for TakuraBid System
Date	October 2025
Status	Draft Version 1.0
Prepared For	TakuraBid Project Stakeholders & Development Team
Reference	SRS_TakuraBid_V1.0_2025
Prepared By	TakuraBid Development Team – B. Machangu, E. Muyambiri, P. Kwaniya, N. Nyekete
Approved By	Project Supervisor: Mrs. Amos
Confidentiality	This document is confidential and intended solely for academic and development purposes under the Software Engineering Project module.

1. Introduction

Purpose

The purpose of this Software Requirements Specification (SRS) is to describe the design, functionality, and performance expectations of the Takura Bid Ride-Hailing Platform, a digital system developed to connect truck drivers with clients who require goods transported efficiently and securely. The platform is designed to simplify the process of booking, tracking, and managing freight transportation services through a user-friendly mobile and web interface.

This SRS provides a complete overview of the system requirements for developers, stakeholders, and users. It outlines what the system will do, how it will perform, and the specific constraints under which it must operate.

Scope

The Takura Bid system will act as a centralized ride-hailing and freight management application. It enables clients to request transportation services, track shipments in real time, and communicate directly with truck drivers. Drivers, on the other hand, can receive trip requests, accept or decline bids, view delivery details, and manage their schedules through the platform.

The main features of the system include:

- Registration and authentication for drivers and clients.
- Real-time ride matching based on proximity, load size, and destination.
- GPS-based route tracking and navigation.
- Secure in-app payment processing and invoice generation.
- Rating and feedback systems for quality control.
- Notifications and trip history management.
- Analytics dashboards for administrators and fleet owners.

The system aims to reduce idle time for truck drivers, improve logistics efficiency, and provide a transparent, secure way to connect transporters with customers.

Definitions and Acronyms

- Driver: A registered truck operator using the platform to find ride or delivery requests.
- Client: A registered user or business requesting freight transport.
- Admin: Platform operator responsible for system maintenance, monitoring, and user management.
- Trip: A delivery or transport request from a client that is accepted by a driver.
- GPS: Global Positioning System used for location tracking.
- API: Application Programming Interface used for integration and data exchange.

References

- IEEE Standard 29148-2018 (Systems and Software Engineering – Life Cycle Processes – Requirements Engineering).
- ISO/IEC 25010:2011 Software Quality Standards.
- Takura Bid System Development Plan and Project Annex-A.

2. Overall Description

Product Perspective

The Takura Bid system is a web and mobile-based application. It will consist of client-facing interfaces (mobile app and website), a driver app for truck operators, and an administrator dashboard for monitoring and analytics. The system will rely on cloud-based infrastructure to ensure scalability and continuous availability.

The system integrates with Google Maps APIs for real-time route mapping and distance estimation, a payment gateway for secure financial transactions, and a notification service for SMS and email alerts.

Intended Audience

The **TakuraBid Project Team** is responsible for designing, coding, and testing the system, ensuring it meets the required specifications and functions smoothly. **Supervisors and lecturers**, including Mrs. Amos, Mr. Duri, and Mr. Manyori, provide technical oversight, constructive feedback, and evaluate the project's progress and quality.

Clients, such as SMEs, corporates, and individuals, are the end users who post shipment requests and track deliveries through the platform. **Drivers**, whether independent or company-affiliated, use the platform to bid for shipments and provide transportation services efficiently.

System administrators manage the platform's configuration, monitor transactions, and handle dispute resolution to maintain smooth operations. Finally, **investors and other stakeholders** review the system's viability and assess its potential for business scalability and growth.

Project Scope

TakuraBid is a digital logistics marketplace designed to bridge the gap between clients (shippers) and drivers (transport providers) in Zimbabwe and beyond. The platform simplifies shipping by providing a transparent, efficient, and user-friendly environment for posting and fulfilling delivery requests. Its core features include:

- **Real-time shipment posting and bidding system:** Clients can post delivery requests instantly, and drivers can competitively bid for jobs, ensuring timely service and fair pricing.

- **AI-based recommendations:** The platform uses intelligent algorithms to match clients with the most suitable drivers, enhancing efficiency and user satisfaction.

- **Instant payment integration:** Payments are seamlessly processed through EcoCash, PayPal, and Stripe, allowing for secure and convenient transactions.

- **KYC-based verification:** Both drivers and clients undergo verification to ensure trust, reliability, and compliance within the platform.

- **Dispute resolution and rating system:** Users can resolve conflicts effectively, and drivers and clients are rated to maintain accountability and service quality.

- **GPS tracking for deliveries:** Real-time tracking allows clients and drivers to monitor shipment locations and progress.

- **Mobile and web applications:** TakuraBid is accessible across devices, allowing users to engage with the system anytime, anywhere.

Out of Scope (Current Phase)

The following features are planned for future phases and are not included in the current development cycle:

- **Blockchain-based transaction logging** (future phase)

- **Advanced predictive analytics**

- **Integration with international carriers**

- **Multilingual interface** (Phase 2)

- **Physical hardware deployment**

Product Functions

The main functions of the system include:

- User Registration and Authentication: Allows both drivers and clients to create accounts using personal or business details.

- Trip Creation and Matching: Clients can post trip requests specifying pickup location, destination, cargo type, and preferred schedule. The system automatically matches available drivers based on proximity, vehicle capacity, and availability.

- Real-Time Tracking: GPS integration allows clients to monitor their cargo and drivers to navigate efficiently.

- Payment and Billing: Supports multiple payment methods including mobile money, debit/credit cards, and electronic invoicing.

- Trip Management: Users can cancel, modify, or reschedule trips with automatic fare adjustment.

- Rating System: After each trip, both driver and client can rate each other to ensure service quality.

- Notifications: The system sends alerts about trip confirmations, cancellations, payments, and delivery completion.

Product Perspective

TakuraBid is a new system designed to transform logistics operations through digitalization and artificial intelligence.
It will serve as a two-sided platform:

- Clients post shipment requests.

- Drivers bid competitively and execute deliveries.

The system integrates:

- AI recommendation APIs for intelligent driver-load matching.
- Payment gateways for secure financial transactions.
- Tracking modules for real-time monitoring.

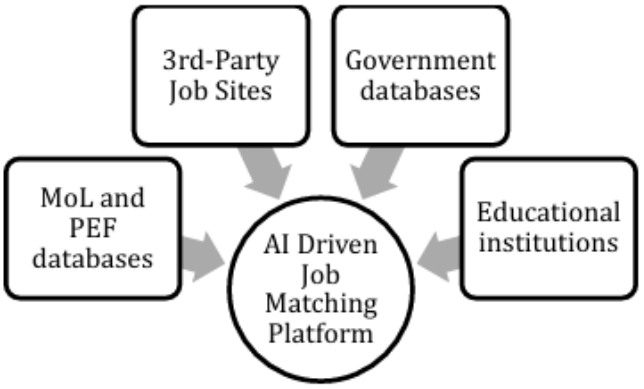
User Characteristics

The primary users include:

- Drivers: Truck owners or operators with valid licenses and registered vehicles.
- Clients: Individuals or companies requiring freight delivery services.
- Administrators: System managers responsible for verifying drivers, handling disputes, and ensuring compliance.

Users are expected to have basic smartphone or computer literacy and reliable internet connectivity.

Context Diagram (Conceptual):



Operating Environment

- Web application: accessible via Chrome, Edge, Firefox.
- Mobile app: Android (Phase 1), iOS (Phase 2).
- Hosting: Cloud-based (AWS / Google Cloud).
- Database: MySQL / PostgreSQL.
- AI Integration: OpenAI API for recommendations.
- Payment Gateways: EcoCash, PayPal, Stripe.

Design Constraints

- The platform must ensure secure data handling and GDPR compliance.
- It must provide 99.9% uptime and real-time performance for ride matching.
- All communications must be encrypted (HTTPS).
- The mobile app must function in offline mode for areas with poor connectivity.

User Documentation

- User Manuals: Client, Driver, and Admin guides.
- Quick Start Guide: Installation and first-time setup.
- Video Tutorials: App usage and safety instructions.
- FAQs & Troubleshooting section on website.

Assumptions:

- Users have reliable internet access.
- Drivers possess smartphones with GPS.
- Payment systems (EcoCash, PayPal) are operational.

Dependencies:

- Integration with payment APIs.
- Availability of OpenAI and Google AI APIs.

Cloud infrastructure availability.

3. System Features

Ride and Trip Management

The system will allow clients to create detailed trip requests that include cargo type, pickup and drop-off addresses, time preferences, and budget range. Drivers will receive available trip notifications in real time, allowing them to bid or accept based on their current status.

Trips can be categorized as immediate, scheduled, or contract-based. Once confirmed, both parties can view the trip details, communicate through in-app messaging, and track the ongoing delivery until completion.

The platform will automatically calculate the fare estimate using distance, vehicle type, and fuel rates. Upon trip completion, the fare will be automatically deducted and transferred to the driver's account after applicable fees.

AI-Based Matching System

Takura Bid will implement an AI-driven matching algorithm that considers driver proximity, vehicle type, route history, and cargo compatibility. The algorithm continuously learns from past data to improve ride recommendations.

The AI will also recommend trips to drivers based on their previous preferences and performance, while suggesting reliable drivers to clients.

Route Optimization and Tracking

The system will use GPS data to provide real-time location tracking and estimated arrival times. Route optimization ensures minimal travel distance, lower fuel usage, and faster delivery.

Drivers can view live traffic updates and alternative route suggestions. Clients can monitor the progress of their goods from pickup to delivery through the map interface.

Payment and Transaction Management

All payments will be processed through integrated gateways supporting local and international methods. The system will automatically generate digital receipts and invoices.

Refunds, cancellations, and penalties will be handled according to predefined policies, and all transactions will be recorded in the database for auditing.

Notifications and Alerts

Takura Bid will provide notifications through SMS, email, and in-app messages.

These include:

- Trip confirmations and cancellations.
- Payment receipts and balance updates.
- Driver arrival alerts and delivery completion notices.

Users can manage their notification preferences in their profile settings.

Ratings and Reviews

After each trip, both clients and drivers will rate the experience using a 5-star system. The average rating will affect future ride recommendations and user credibility. Feedback comments will be analyzed to identify trends in service quality.

Administration and Analytics

Administrators will have full visibility into trip logs, payment records, and user activity. The dashboard will display system performance metrics such as total trips, active drivers, and daily revenue.

Reports can be exported in PDF, Excel, or CSV formats for decision-making.

3.2. Job Posting and Management

3.2.1. Job Creation and Publishing

Requirement

a) The system SHALL offer a structured form for job postings that allows employers to create job advertisements.

b) The system SHALL offer a structured form for job postings that enables employers to add new job opportunities with the following fields:

● Job title, a brief summary, and required skills (with options for file upload or copy/paste input)

● Type of contract (e.g., full-time, part-time, training, project-based)

● Necessary education level

● Application deadline with an optional feature to auto-close

● Selection of work format (On-site with location, Remote, or Hybrid)

● Gender, number of positions, etc.

● Required languages and their proficiency levels

● Link to the job listing if available.

● Additional detailed features for the job posting will be provided.

- The system SHALL accommodate standard job classifications, skills, and qualifications to ensure precise matching. To fulfill this requirement, the system MUST guarantee that all input data adheres to the Schema.org JobPosting standard for semantic alignment and the integrity of structured data.

c) The system SHALL enable employers to oversee their job listings by modifying job information, prolonging application deadlines, and determining the visibility of job

postings (public, private, or targeted) as needed.

d) The system SHALL facilitate job posting expiration and renewal procedures.

3.2.2. Job Search and Filtering

Requirement

a) The platform SHALL offer a detailed search interface for job seekers to discover relevant job opportunities.

b) The search interface SHALL accommodate both basic (searching by a single keyword) and advanced search features based on attributes defined by employers.

c) The system SHALL incorporate semantic search functions that grasp the intent behind search queries.

d) The platform SHALL present a search engine with enhanced filtering options, enabling job seekers to look for positions using the following criteria:

- Keywords (text)
- Location
- Salary range
- Employment type
- Date posted
- Application deadline
- Job-specific requirements (e.g., skills, education, training, years of experience)
- Company sector/industry (e.g., IT, Health, NGO, Construction)

e) The system SHALL show search results ranked by relevance/skills matching and provide sorting options. Moreover, once users log in, the system SHALL display suggested jobs based on results from the AI matching engine.

f) The platform SHALL feature a favorites option (e.g., heart button) for users to bookmark

jobs that interest them.

g) The system SHALL allow users to save searches and receive notifications for new matching job opportunities.

3.2.3. Job Interaction Process

Requirement

- The system SHALL allow job seekers to bookmark and save jobs by adding them to an “Interested List ”, and to save custom search filters for future

3.2.4. Job Status Tracking

Requirement

a) The system SHALL provide employers with tools to manage the status of job postings.

b) The system SHALL support the following job posting statuses: draft, active, paused, expired, and archived.

c) The system SHALL maintain a history of status changes for audit and reporting purposes.

3.3. AI-Driven Matching System

3.3.1. A Vector-Based (Skills-Based and Behavior) Matching Algorithms

Requirement

a) The system SHALL implement AI-driven matching algorithms to connect job seekers with relevant job opportunities.

b) The system SHALL use a scoring algorithm to calculate a match score between job seekers and job postings minimally based on the following criteria:

- Skill overlap
- Education match
- Training match
- Location match
- Experience range
- Salary expectation range

c) The system SHALL generate shortlists of top-matching candidates (e.g., top 100) for each job posting to assist employers in the selection process.

d) The system SHALL use natural language processing (NLP) to understand the semantic meaning of job descriptions and resumes beyond keyword matching.

e) The system SHALL perform keyword and semantic analysis on posted job descriptions to identify key attributes such as required skills, experience levels, and job categories.

f) The system SHALL allow administrators to define and adjust a minimum match threshold (e.g., 60%) that determines which job matches are displayed to users.

g) The system SHALL rank/order job postings for each job seeker based on their individual match percentage, displaying the most relevant opportunities first.

h) The system SHALL support two-way matching, where job seekers receive ranked job opportunities based on match percentage, and employers receive reverse matches with ranked lists of suitable job seekers for their postings.

i) The system SHALL allow configuration of matching parameters to adjust the importance of different factors.

j) The system SHALL implement AI-powered resume parsing to extract structured information from uploaded documents.

k) The system SHALL extract the following information from resumes: personal details, contact information, education history, work experience, skills, certifications, and achievements.

- The system SHALL support multiple languages in resume parsing, with primary focus on Arabic and English.
- The system SHALL identify and standardize skills mentioned in resumes to facilitate matching.
- The system SHALL provide confidence scores for extracted information and highlight areas that may need manual verification.
- The system SHALL allow job seekers to review and correct parsed Information.

3.3.2. Job Recommendation Engine

Requirement

- The system SHALL provide personalized job recommendations for job seekers based on their profile, preferences, and platform activity.
- The system SHALL recommend suitable job seekers to employers based on job requirements and matching criteria.

- The system SHALL use collaborative filtering to recommend jobs based on similar user behaviors and preferences.
- The system SHALL incorporate content-based filtering to recommend jobs similar to those the user has shown interest in.
- The system SHALL consider location, salary expectations, and work arrangement preferences in recommendations.

3.3.3. Candidate Recommendation for Employers

Requirement

- The system SHALL provide employers with candidate recommendations for their job postings.
- The system SHALL rank candidates based on match quality, highlighting the strengths and potential gaps of each candidate.
- The system SHALL allow employers to set minimum qualification thresholds for automatic candidate filtering.
- The system SHALL allow employers to search the candidate database using advanced filtering options.
- The system SHALL respect candidate privacy settings when making recommendations to employers.
- The system SHALL provide insights on candidate availability, salary expectations, and potential fit.
- The system SHALL allow employers to save promising candidates to talent pools for future opportunities (add to shortlist).

3.4. Integration with External Systems

3.4.1. External Job Site Integration

Requirement

- The system SHALL support integration with external job sites based on recommendations by MoL and PEF.
- The system SHOULD provide APIs for real-time job data synchronization with external job sites.
- The system SHOULD implement data mapping and transformation to standardize job postings from different sources.

- The system SHOULD support both pull (importing jobs from external sites) and push (exporting jobs to external sites) integration models.
- The system SHOULD maintain synchronization logs and provide error handling for failed integrations.
- The system SHOULD support scheduled and on-demand synchronization options.
- The system SHOULD provide a dashboard to monitor integration status and data flow with external job sites.

3.4.2. Government Database Integration

Requirement

- The system SHOULD integrate with relevant government databases for data verification and enrichment.
- The system SHOULD support integration with the MoL and PEF existing databases and systems.

- The system SHOULD support verification of educational credentials through integration with educational institution databases.
- The system SHOULD provide options for identity verification through government ID systems.
- The system SHOULD maintain audit trails of all data exchanges with government systems.
- The system SHOULD respect data privacy regulations when accessing and using government data.

3.4.3. API Framework

Requirement

- The system SHOULD provide a comprehensive API framework for integration with external systems.
- The system SHOULD implement RESTful APIs with JSON data format as the primary integration method.
- The system SHOULD provide detailed API documentation with examples and testing tools.
- The system SHOULD implement OAuth 2.0 for API authentication and authorization.
- The system SHOULD support API versioning to ensure backward Compatibility.

3.5. Reporting and Analytics

3.5.1. User Activity Monitoring

Requirement

- The system SHALL track and record user activities for analysis and reporting.
- The system SHALL monitor job seeker activities including profile views, job searches, applications, and interactions.
- The system SHALL TRACK employer activities including job postings, candidate searches.
- The system SHALL implement data retention policies for activity logs in compliance with regulations.

- MOL Life dashboard should include current logins, number of current logins and last sessions, with links to these users profiles

3.5.2. Employment Statistics

Requirement

- The system SHOULD generate employment statistics and labor market insights from platform data.
- The system SHOULD track key metrics including job posting trends, application rates, hiring rates, and time-to-fill.
- The system SHOULD provide industry-specific analytics on job market demand and supply.
- The system SHOULD analyze skill demand trends to identify emerging requirements and skill gaps.

- The system SHOULD generate geographic distribution reports for jobs and candidates.
- The system SHOULD provide salary range analytics by industry, position, and location.
- The system SHOULD track employment outcomes and career progression where data is available.
- The system SHOULD generate periodic labor market reports for government Stakeholders.

3.5.3. System Performance Metrics

Requirement

- The system SHOULD monitor and report on system performance metrics.
- The system SHOULD track matching algorithm performance including accuracy, precision, recall, and user satisfaction.
- The system SHOULD monitor system usage patterns including peak times, popular features, and user engagement.
- The system SHOULD track technical performance metrics including response times, resource utilization, and error rates.

- The system SHOULD provide dashboards for administrators to monitor system health and performance.
- The system SHOULD generate alerts for performance issues or anomalies.
- The system SHOULD maintain historical performance data for trend analysis and capacity planning.

3.5.4. Custom Report Generation

Requirement

- The system SHOULD provide a flexible reporting framework for generating custom reports through seamless integration with existing and available reporting tools such as Microsoft Power BI.
- The system SHOULD allow administrators to define report templates with configurable parameters.
- The system SHOULD support various report formats including tabular data, charts, and visualizations.
- The system SHOULD allow scheduling of recurring reports with automated distribution.
- The system SHOULD support export of reports in common formats (PDF, Excel, CSV).
- The system SHOULD provide a report builder interface for users with appropriate permissions.
- The system SHOULD maintain a library of saved reports for quick access.
- The system SHOULD implement access controls to ensure users can only view reports appropriate to their role.

3.6. Notification System

3.6.1. Email Notifications

Requirement

- The system SHALL send email notifications for important events and updates.
- The system SHALL support customizable email templates with dynamic

content.

- The system SHALL allow users to configure their email notification preferences.
- The system SHALL support both immediate and digest email notifications.
- The system SHALL log all emails notifications.
- The system SHALL comply with anti-spam regulations and best practices.

3.6.2. In-App Notifications

Requirement

- The system SHALL provide an in-app notification center for users.
- The system SHALL display real-time notifications for important events and updates.
- The system SHALL provide weekly job recommendations to job seekers based on their search history and profile information.
- The system SHALL maintain a notification history for users to review past notifications.
- The system SHALL allow users to configure their in-app notification preferences.
- The system SHALL support different notification types with appropriate visual indicators.
- The system SHALL provide notification management tools for users to mark as read, delete, or take action on notifications.

3.6.3. SMS Notifications

Requirement

- The system SHALL provide SMS notifications for critical updates and time sensitive information (byt the integration with SMS service providers (gateway)). Such as welcoming SMS with a code for user registration, employer registration, and SMS for resetting the password.
- The system SHALL allow users to opt-in to SMS notifications and provide their mobile number.
- The system SHALL limit SMS notifications to essential communications to avoid overwhelming users.

- The system SHALL track SMS delivery status for monitoring and troubleshooting. + loging as sms messages
- The system SHALL comply with telecommunications regulations regarding SMS messaging.

3.7. Content Management

3.7.1. News and Updates

Requirement

- The system SHALL provide a content management system (CMS) for publishing news and updates.
- The system SHALL allow administrators to create, edit, and publish articles and announcements.

- The system SHALL support rich text formatting, images, and embedded

media in content.

- The system SHALL provide content categorization and tagging for organization.
- The system SHALL display relevant news and updates on user dashboards based on their profile.
- The system SHALL maintain an archive of past news and updates with search Functionality.

3.7.2. FAQ and Help Center

Requirement

- The system SHALL provide a comprehensive FAQ and help center. In addition, list of laws, regulations and contract types regulations.
- The system SHALL organize help content by topic and user role for easy navigation.
- The system SHALL implement a search function for help content.
- The system SHALL provide context-sensitive help throughout the platform.
- The system SHALL allow administrators to update help content as the system evolves.
- The system SHALL collect user feedback on help content effectiveness.
- The system SHALL provide guided tours and tutorials for new users.
- The system SHALL support multimedia help content including videos and interactive guides.

4. External Interface Requirements

4.1. User Interfaces

The system shall provide user interfaces for the following user types:

1) Job Seeker

- Registration and profile management interface

- Job search and application interface

- Dashboard for tracking applications and recommendations

Notification and messaging interface

2) Employers

- Registration and company profile management interface

- Job posting and management interface

- Candidate search and evaluation interface

- Recruitment workflow management interface

3) External Job Sites

- Integration configuration interface

- Data mapping and synchronization interface

- Monitoring and reporting interface

4) Administrators

- User\roles\permissions management interface

- System configuration interface

- Content management interface

- Reporting and analytics interface

5) Guest User

- The system SHALL allow unregistered (guest) users to browse available job listings

using

basic filters such as location, sector, and date posted.

- The system SHALL prompt guest users to register or log in when they attempt to apply for a job, save listings, or subscribe to notifications.
- The system SHALL ensure that privacy policies and cookie consent banners are clear, guest friendly, and fully compliant with GDPR and applicable local data protection laws, providing transparency in data collection and usage.

All user interfaces shall adhere to the following requirements:

- The system SHALL have a user-friendly UX/UI dynamic interface that showcases the latest job postings, featured opportunities, and relevant sector-based news or career advice, all presented in a visually engaging and user-friendly layout
- Responsive design for desktop and mobile devices
- Multilingual support (Shona and English) and different AI functions and techniques for both Shona and English to be utilized
- Implement Web Content Accessibility Guidelines (WCAG 2.1 or later) to ensure accessibility and inclusivity for users with disabilities, including support for screen readers, appropriate color contrast, and keyboard navigation.
- Consistent navigation and design patterns
- Display dynamic homepage content, including the latest job postings, featured opportunities, and relevant sector-based news or career advice, presented in an engaging and easy-to-navigate layout.
- Context-sensitive help and guidance
- All public pages must be SEO-optimized

4.2. Hardware Interfaces

The system shall interface with the following hardware components:

1) Server Infrastructure

- Application servers

- Database servers

- Storage systems

- Backup systems

2) Network Infrastructure

- Load balancers

- Firewalls

- Network monitoring devices

3) Client Devices

- Desktop computers

- Laptops

- Tablets

- Smartphones

The system shall be designed to minimize hardware dependencies and support a wide range of

client devices and configurations.

4.3. Software Interfaces

The system shall interface with the following software systems:

1) External Job Sites

- API integration with at least five (5) external job sites

- Data exchange protocols and formats

- Authentication and security mechanisms

2) Government Databases

- Integration with MoL and PEF existing systems

- Integration with educational institution databases

- Integration with identity verification systems

3) Email and SMS Gateways

- Integration with email service providers

- Integration with SMS gateways

4) Analytics and Reporting Tools

- Integration with data visualization tools

- Export capabilities for external analysis

- All software interfaces shall be documented with detailed specifications, including data formats, protocols, and security requirements.

4.4. Communications Interfaces

The system shall support the following communication interfaces:

1) Network Protocols

- HTTP/HTTPS for web access

- WebSockets for real-time notifications

- SMTP for email communications

- SMS protocols for mobile notifications

2) API Communications

- RESTful/SOAP API for external integrations

- J-SON data format for data exchange

- OAuth 2.0 for authentication

3) Data Exchange Formats

- JSON for API data exchange

- XML for legacy system integration where required

- CSV for data import/export

- All communications shall be secured using appropriate encryption and authentication mechanisms.

Nonfunctional Requirements

The system must deliver a fast, reliable, and secure user experience. It should support at least 1,000 concurrent users without degradation. Passwords and user data must be encrypted, and the codebase must follow OWASP security standards.

The application will be maintainable, scalable, and capable of handling an increased number of trips as user adoption grows.

Uptime must remain above 99.9% per month, excluding scheduled maintenance.

The system must recover from crashes within five minutes using automated backups and failover servers.

System Models and Diagrams

The system will include:

- A Context Diagram showing how users, APIs, and databases interact.
- A Use Case Diagram representing the relationship between clients, drivers, and administrators.
- A Data Flow Diagram (Level 0 and Level 1) outlining data movement between modules.
- A System Architecture Diagram depicting all components, including backend services and communication layers.

Other Requirements

The platform must comply with transport and e-commerce regulations, including vehicle verification and driver background checks.

It will also maintain data privacy according to GDPR standards and allow data portability for users wishing to export their trip history.

5. Non-Functional Requirements – TakuraBid Freight Matchmaking Platform

This section specifies the non-functional requirements for the TakuraBid Platform. These define the quality attributes and constraints that the system must satisfy to meet stakeholder expectations.

5.1. Performance Requirements

5.1.1. Response Time

ID	Requirement
NFR-01	The system SHALL provide page load times of less than 3 seconds for standard operations under normal load conditions.
NFR-02	The system SHALL return load search and driver bid results within 2 seconds for standard queries.
NFR-03	The system SHALL provide recommendation results (driver or price suggestions) from OpenAI APIs within 5 seconds.
NFR-04	The system SHALL process bulk operations (e.g., multiple load updates or bid evaluations) within 2 minutes for standard workloads.
NFR-05	The system SHALL maintain a maximum of 50% response time degradation during peak load conditions.

5.1.2. Throughput

ID	Requirement
NFR-06	The system SHALL support at least 1,000 concurrent users during normal operations.
NFR-07	The system SHALL support up to 2,000 concurrent users during peak periods (target growth).
NFR-08	The system SHALL process at least 100 bid transactions per minute during high activity.
NFR-09	The system SHALL support at least 500 new load postings per day.
NFR-10	The system SHALL support at least 300 new driver registrations and 200 client registrations per day.

5.1.3. Resource Utilization

ID	Requirement
NFR-11	The system SHALL utilize no more than 80% CPU during normal operations.
NFR-12	The system SHALL utilize no more than 80% memory during normal operations.
NFR-13	The system SHALL require no more than 3TB of storage in the first year of operation, with growth planning for future years.
NFR-14	The system SHALL optimize PostgreSQL queries to minimize I/O and improve response time.
NFR-15	The system SHALL implement server-side caching (Redis or equivalent) to reduce repetitive query load.

5.1.4. Scalability

ID	Requirement
NFR-16	The system SHALL scale horizontally by adding more AWS ECS containers to handle increased traffic.
NFR-17	The system SHALL scale vertically by increasing resources on existing AWS instances.
NFR-18	The system SHALL support 10,000 registered users without performance degradation.
NFR-19	The system SHALL support a minimum of 5,000 registered drivers without degradation
NFR-20	The system SHALL support at least 50,000 load records in the database efficiently.
NFR-21	The system SHALL accommodate a 100% annual growth in transaction volume and active users for the first three years.

5.2. Security Requirements

5.2.1. Authentication and Authorization

ID	Requirement
NFR-22	The system SHALL enforce strong password policies for all users.
NFR-23	The system SHALL implement role-based access control (RBAC) for drivers, clients, and administrators.
NFR-24	The system SHALL log all access and authentication events.
NFR-25	The system SHALL lock accounts after 5 consecutive failed login attempts.
NFR-26	The system SHALL implement secure session management with appropriate timeouts.
NFR-27	The system SHALL support third-party authentication (e.g., Google OAuth) where applicable for user convenience.

5.2.2. Data Protection

ID	Requirement
NFR-28	The system SHALL encrypt all sensitive data at rest using AES-256 encryption.
NFR-29	The system SHALL encrypt all data in transit using TLS 1.3 or higher.
NFR-30	The system SHALL securely store API keys (e.g., OpenAI, Google Maps) using AWS Secrets Manager.
NFR-31	The system SHALL provide mechanisms for secure data deletion when users request account closure.
NFR-32	The system SHALL isolate development, testing, and production environments.

5.2.3. Privacy and Compliance

ID	Requirement
NFR-33	The system SHALL comply with Zimbabwean data privacy regulations and incorporate GDPR principles where applicable.
NFR-34	The system SHALL allow users to view and delete personal data upon request.
NFR-35	The system SHALL maintain audit trails for all critical data modifications.
NFR-36	The system SHALL collect only necessary data to operate core functions (data minimization principle).
NFR-37	The system SHALL display clear privacy notices and consent messages during registration and data collection.
NFR-38	The system SHALL define and enforce data retention policies in line with operational needs and regulations.

5.2.4. Security Monitoring and Incident Response

ID	Requirement
NFR-39	The system SHALL log all security-related events using AWS CloudWatch or equivalent.
NFR-40	The system SHALL implement real-time alerting for suspicious or unauthorized activities.
NFR-41	The system SHALL include intrusion detection mechanisms in its AWS environment.
NFR-42	The system SHALL perform regular vulnerability scans and penetration tests.
NFR-43	The system SHALL maintain a documented incident response plan.
NFR-44	The system SHALL apply rate limiting to prevent brute-force or denial-of-service attacks.
NFR-45	The system SHALL include automated patch and update management for all components.

5.3. Reliability and Availability

5.3.1. Availability

ID	Requirement
NFR-46	The system SHALL maintain 99.9% uptime under normal operating conditions.
NFR-47	The system SHALL schedule maintenance during off-peak hours.
NFR-48	The system SHALL provide advance notifications of maintenance events.
NFR-49	The system SHALL implement multi-AZ high availability architecture to prevent single points of failure.

.3.2. Fault Tolerance

ID	Requirement
NFR-50	The system SHALL degrade gracefully under component failures (e.g., one service down).
NFR-51	The system SHALL enable database replication for redundancy.
NFR-52	The system SHALL implement load balancing to distribute traffic evenly.
NFR-53	The system SHALL automatically recover from common AWS service failures.
NFR-54	The system SHALL apply circuit breaker patterns for API dependencies (OpenAI, Google Maps).

5.3.3. Disaster Recovery

ID	Requirement
NFR-55	The system SHALL maintain daily incremental and weekly full backups.
NFR-56	The system SHALL store backups in separate AWS regions for safety.
NFR-57	The system SHALL define a Recovery Time Objective (RTO) of 4 hours and Recovery Point Objective (RPO) of 1 hour.
NFR-58	The system SHALL maintain and periodically test a disaster recovery plan.

5.3.4. Error Handling

ID	Requirement
NFR-59	The system SHALL maintain daily incremental and weekly full backups.
NFR-60	The system SHALL store backups in separate AWS regions for safety.
NFR-61	The system SHALL define a Recovery Time Objective (RTO) of 4 hours and Recovery Point Objective (RPO) of 1 hour.
NFR-62	The system SHALL maintain and periodically test a disaster recovery plan.
NFR-63	The system SHALL maintain stability when encountering unexpected inputs.

5.4. Usability and Accessibility

5.4.1. User Interface

ID	Requirement
NFR-64	The system SHALL provide an intuitive and simple interface for both drivers and clients.
NFR-65	The system SHALL use responsive design to work on mobile phones, tablets, and desktops.
NFR-66	The system SHALL provide easy navigation to post loads, browse jobs, and place bids.
NFR-67	The system SHALL provide immediate feedback for user actions (e.g., bid submission confirmation).
NFR-68	The system SHALL minimize the number of steps to complete actions (e.g., posting a load or placing a bid).
NFR-69	The system SHALL include context-based help tips for non-tech-savvy users.

5.4.2. Accessibility

ID	Requirement
NFR-70	The system SHALL comply with **WCAG 2.1 Level AA** accessibility standards.
NFR-71	The system SHALL support screen readers and basic assistive technologies.
NFR-72	The system SHALL allow full keyboard navigation for all functions.
NFR-73	The system SHALL maintain good color contrast for text and buttons.
NFR-74	The system SHALL provide text alternatives for icons or non-text content.

5.4.3. User Experience

ID	Requirement
NFR-75	The system SHALL provide a personalized experience (e.g., showing preferred routes or frequent clients/drivers).
NFR-76	The system SHALL include a clear onboarding process for new users.
NFR-77	The system SHALL incorporate user feedback to improve usability.
NFR-78	The system SHALL support users with different technical skill levels.
NFR-79	The system SHALL minimize user cognitive load by using clear icons, visuals, and short text.t.
NFR-80	The system SHALL use smart defaults (e.g., auto-fill frequent pickup addresses).

5.5. Maintainability

ID	Requirement
NFR-81	The system SHALL use a modular architecture (React frontend, Django backend, PostgreSQL DB).
NFR-82	The system SHALL follow consistent coding standards and naming conventions.
NFR-83	The system SHALL include comprehensive developer documentation.
NFR-84	The system SHALL implement logging and monitoring via AWS CloudWatch.
NFR-85	The system SHALL support configuration changes via environment variables (no code edits).
NFR-86	The system SHALL include automated testing for core components.
NFR-87	The system SHALL use GitHub version control for all code artifacts.

5.6. Documentation

ID	Requirement
NFR-88	The system SHALL provide comprehensive user documentation for all user roles.
NFR-89	The system SHALL provide technical documentation for system administrators and developers.
NFR-90	The system SHALL maintain up-to-date system architecture and design documentation.
NFR-91	The system SHALL provide troubleshooting guides and known issue documentation.

5.7. Cultural and Political Requirements

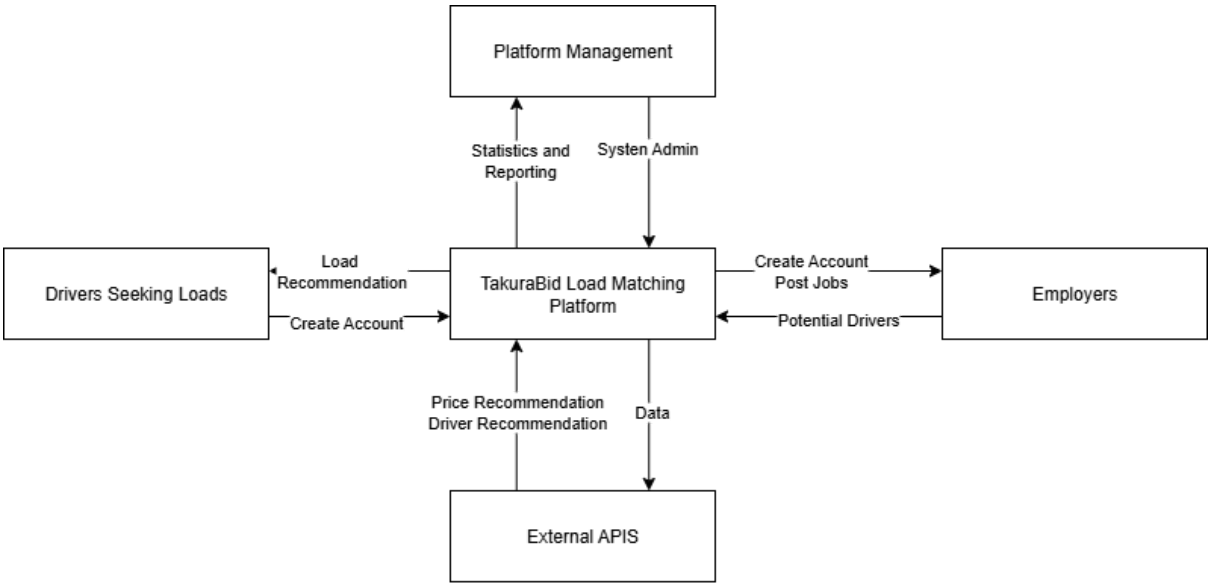
5.7.1. Cultural Considerations

ID	Requirement
NFR-92	The system SHALL respect cultural norms and sensitivities in the Zimbabwean context.
NFR-93	The system SHALL use appropriate terminology and language for the local context.
NFR-94	The system SHALL support local date and time formats
NFR-95	The system SHALL consider gender sensitivities in user interfaces and communications.

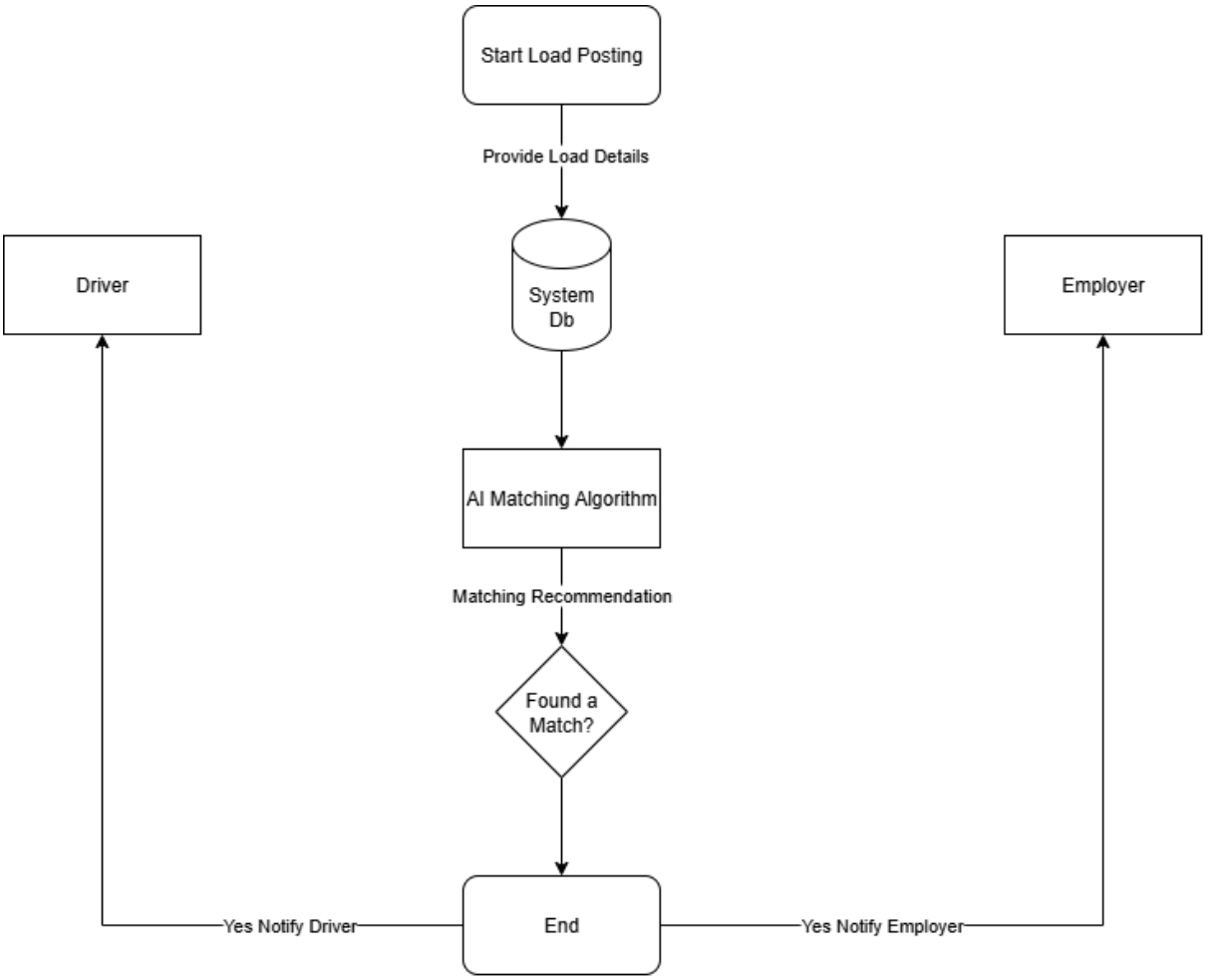
5.7.2. Political Considerations

ID	Requirement
NFR-96	The system SHALL use politically neutral terminology in system interfaces and documentation.
NFR-97	The system SHALL respect the political sensitivities of the region in geographic references and maps.
NFR-98	The system SHALL implement appropriate content moderation policies for politically sensitive content.

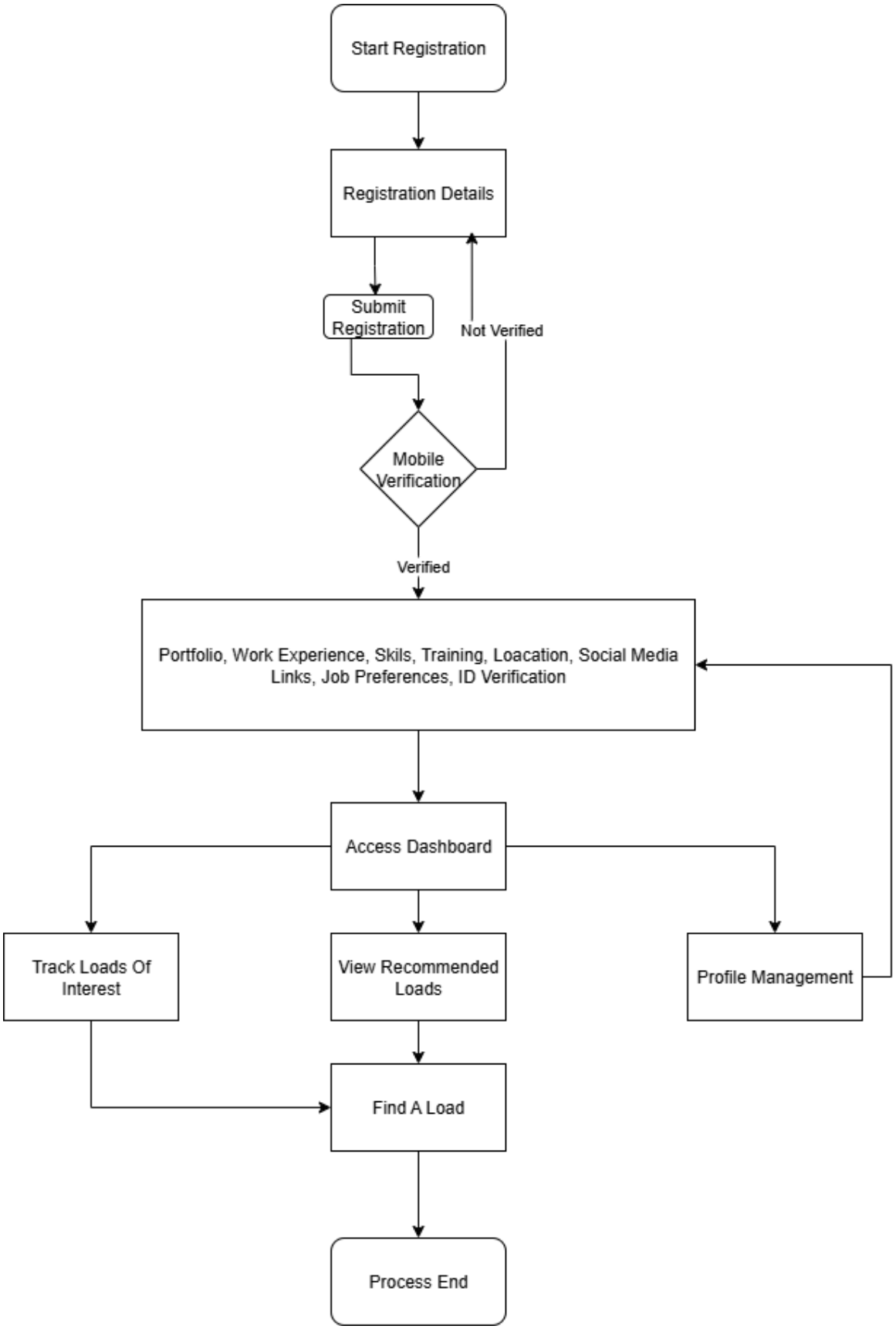
6. Context Diagram - System boundaries and external entities (Level 0)



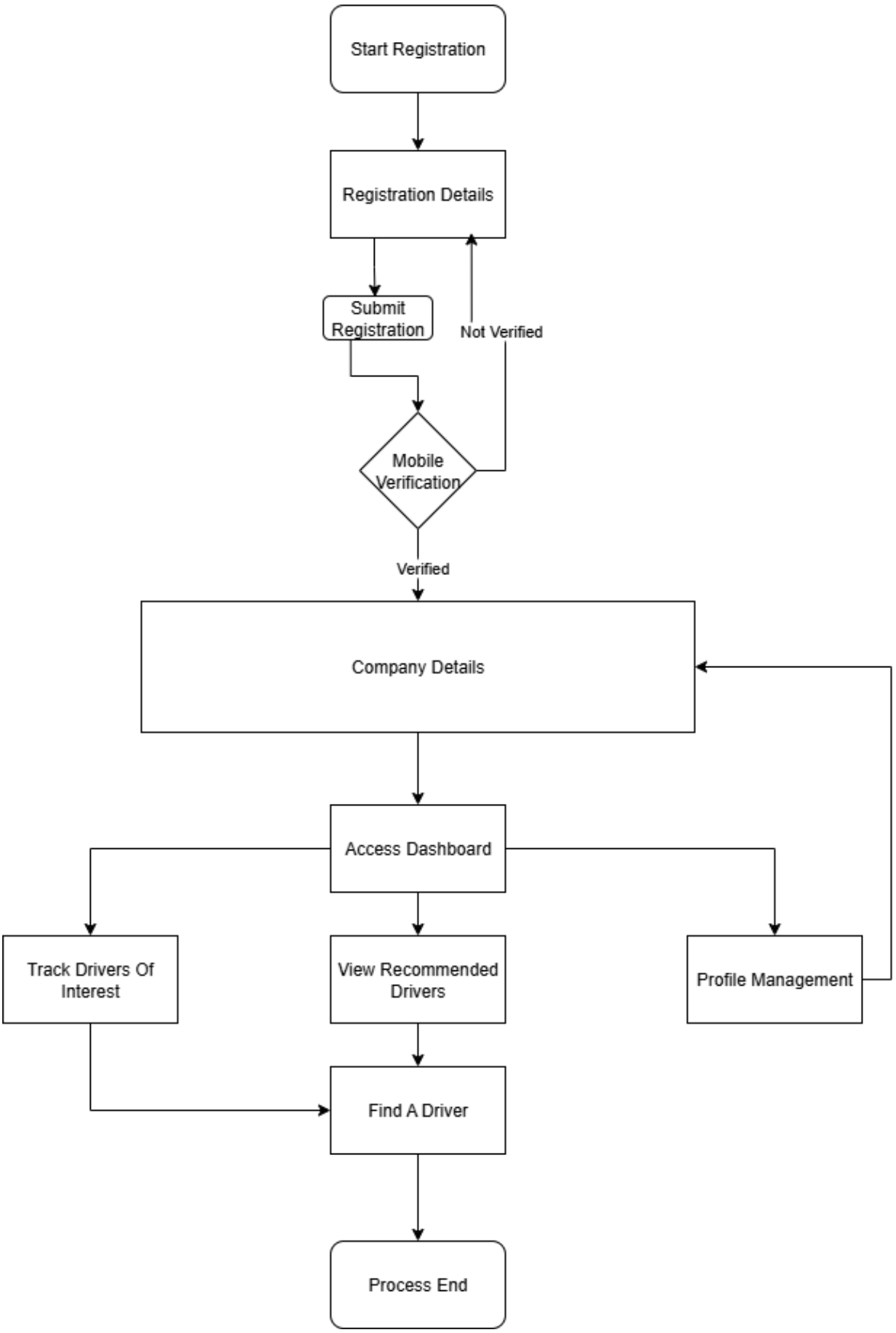
Load posting and application process


Start Load Posting
Provide Load Details
System Db
Driver
Employer
AI Matching Algorithm
Matching Recommendation
Found a Match?
End
Yes Notify Driver
Yes Notify Employer

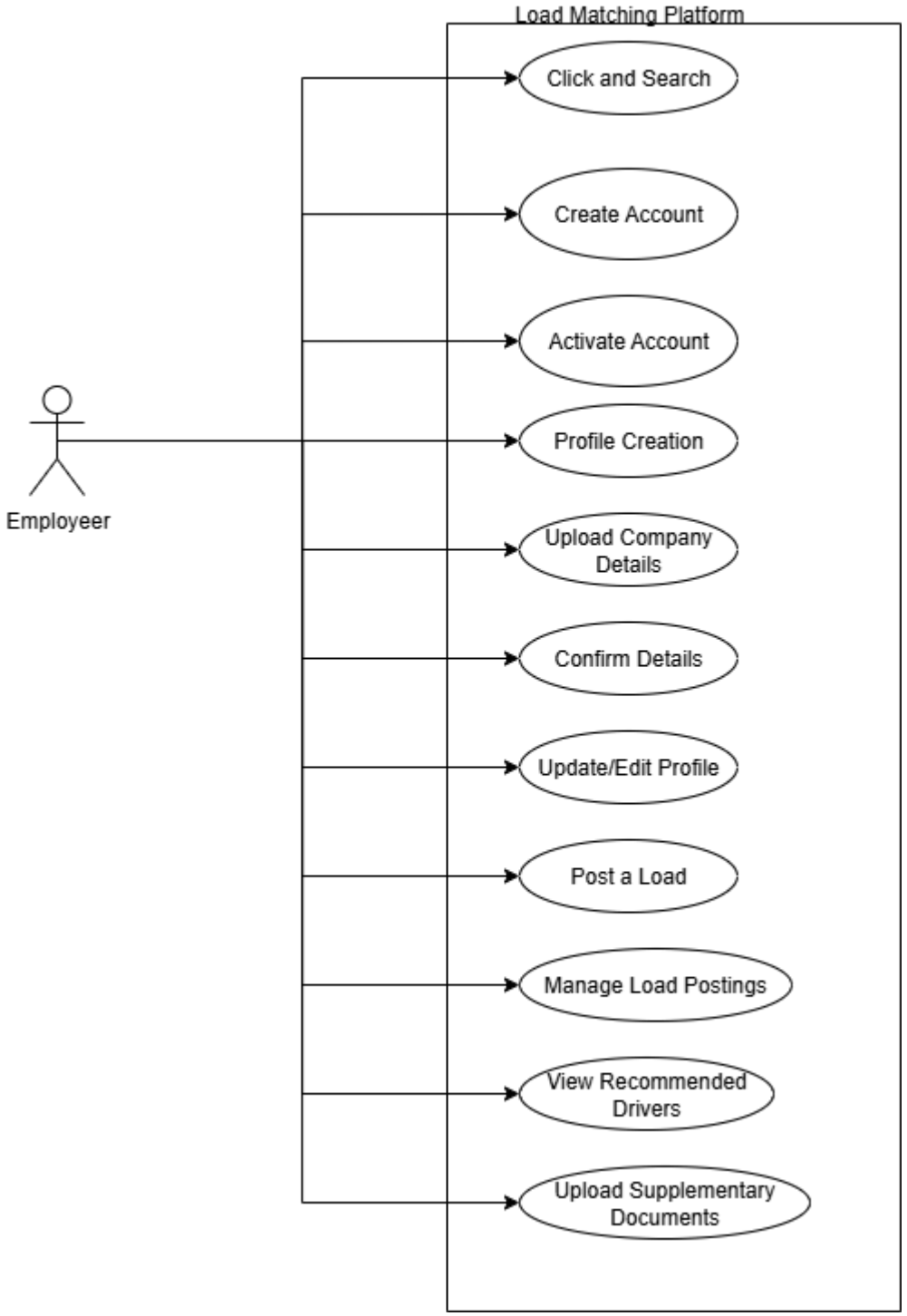
Load seeker registration and profile management



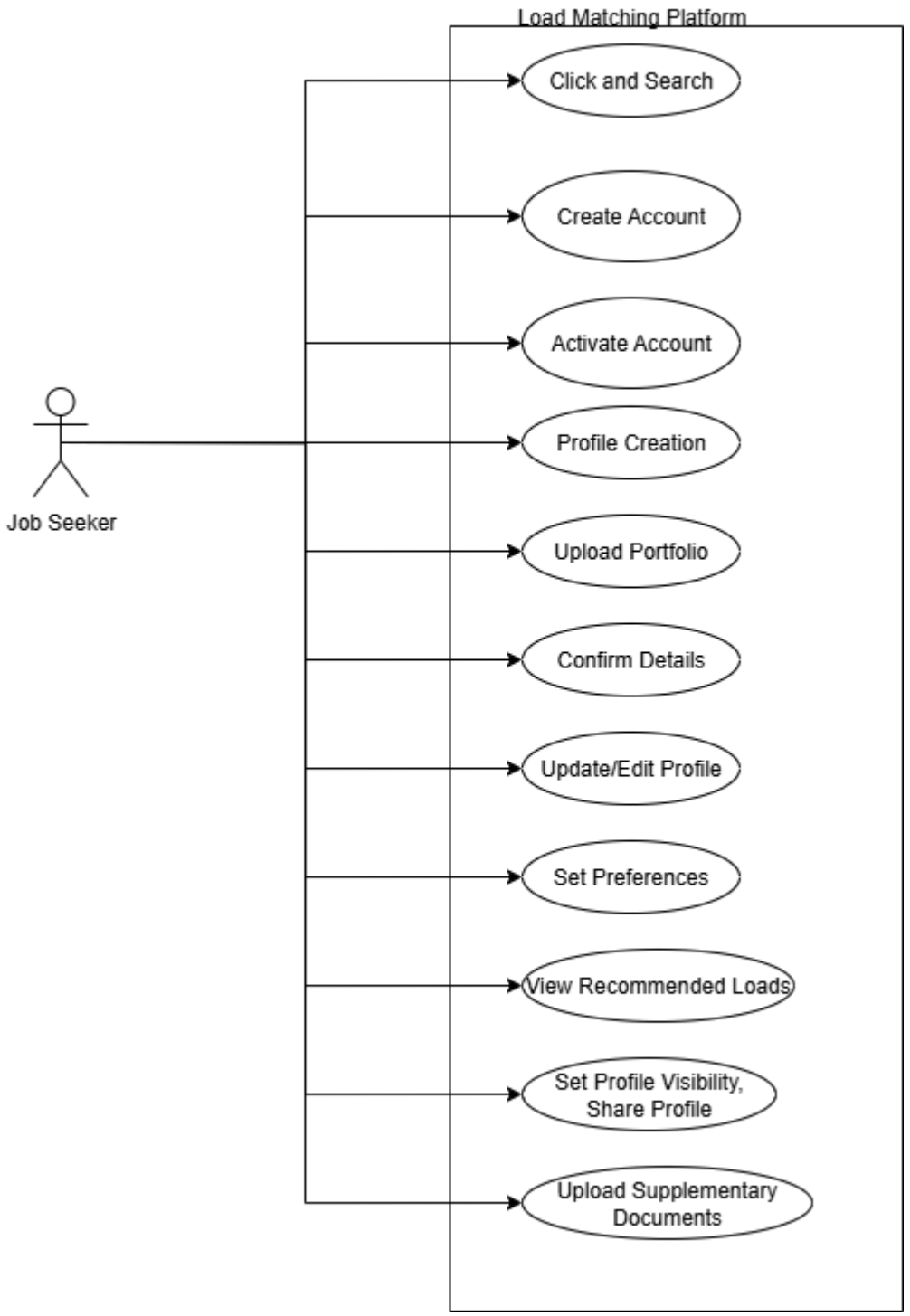
Employer registration and profile management


Start Registration
Registration Details
Submit Registration
Not Verified
Mobile Verification
Verified
Company Details
Access Dashboard
Track Drivers Of Interest
View Recommended Drivers
Profile Management
Find A Driver
Process End

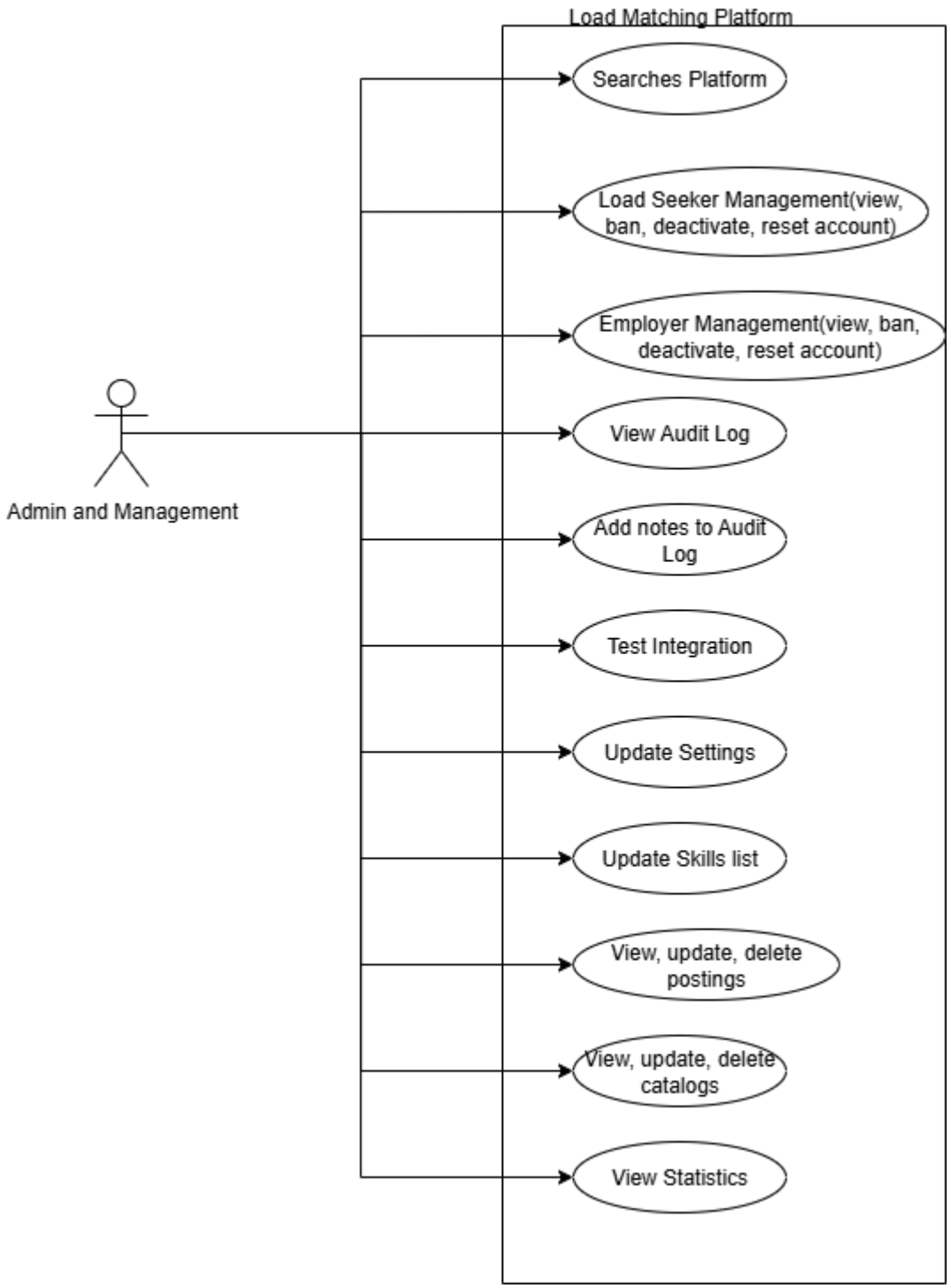
Employers use case diagram



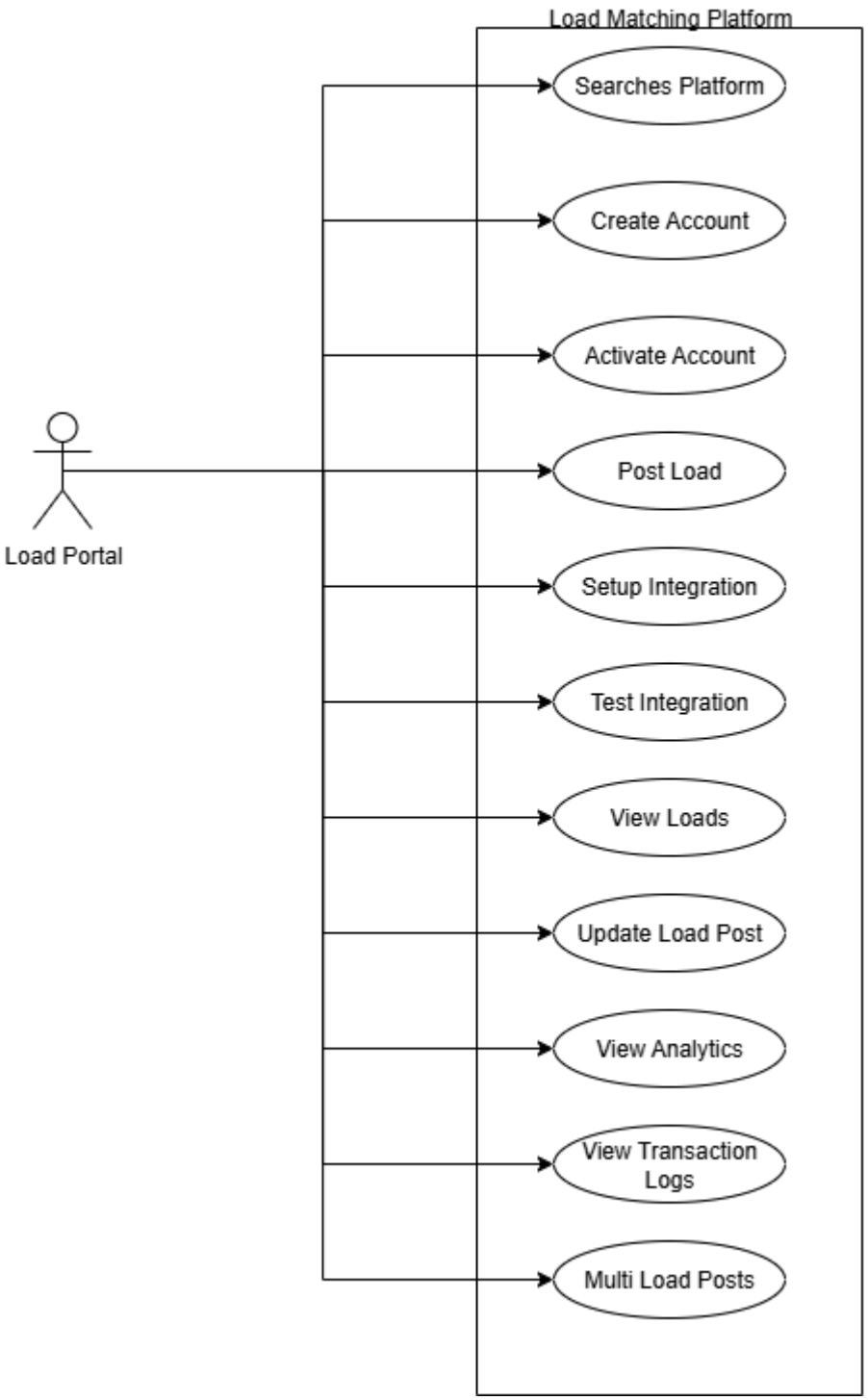
Job seekers use case diagram



Administration Management use case diagram



Load portals use case diagram


Load Matching Platform
Searches Platform
Create Account
Activate Account
Post Load
Setup Integration
Test Integration
View Loads
Update Load Post
View Analytics
View Transaction Logs
Multi Load Posts
Load Portal

6.5. Appendix B: Open Issues & Questions

This appendix tracks unresolved issues and key questions to be addressed during the platform development:

1. **Integration Requirements**

- Comprehensive specifications for the external job/marketplace sites to be integrated
- API documentation, authentication, and access details

2. **Hosting Strategy**

- Finalized choice of hosting model (cloud, on-premises, or hybrid)
- Infrastructure specifications and technical requirements

3. **Data Migration Plan**

- Complete inventory of data slated for migration
- Data quality evaluation and cleansing methodology

4. **AI Matching Algorithm**

- Selection of algorithms for load and carrier matching
- Identification of training data requirements and sources

Note: This list will be continually updated as the project progresses.

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